

GROUP PROJECT GUIDELINE

TOPIC 1: AI IN AGRICULTURE

OVERVIEW

Course: TMA Advanced AI Training

Type: Group Projects (10-15 members per group)

Objectives: This project aims to provide learners with hands-on experience in developing an AI-powered system for agriculture. By utilizing publicly available datasets of plant and leaf images, learners will explore the full AI development pipeline—from data acquisition, preprocessing, and augmentation, to designing, training, and evaluating deep learning models such as Convolutional Neural Networks (CNNs). The project will include performance evaluation using metrics like accuracy, precision, recall, and F1-score, and culminate in deploying the solution through a user-friendly interface or mobile-ready application. The ultimate goal is to create a practical tool that can help farmers identify crop diseases early, reduce yield losses, and support more sustainable and efficient agricultural practices.

DATASET

Learners may either **use a dataset of their choice** or select from one of the recommended datasets listed below:

No.	Name	Source	Description
1	PlantVillage dataset	Link	Dataset of diseased plant leaf images and corresponding labels
2	Multi-class Weather dataset for Image Classification	Link	This dataset contains images of weather suitable for weather analysis
3	Diseases on Citrus Fruit and Leaves	Link	The dataset contains an image gallery of healthy and unhealthy citrus fruits and leaves
4	FAOSTAT dataset	Link	This dataset of various statistics from Food and Agriculture Organization of the UN (FAO-UN)
5	GBIF dataset	Link	International network and data infrastructure funded by the world's governments and aimed at providing access to data about all types of life on Earth

GUIDELINES

Each group will submit 1 report with the following constraints:

- Report must include reference/citation of the sources being used
- Visualization can be included in the report but must not exceed 50% of the total space
- Report can range from 10 to 15 pages

REPORT GUIDELINE

Your final report should be structured according to the **CRISP-DM (Cross-Industry Standard Process for Data Mining)** standard ([Link](#)) and should include the following sections:

1. Business Understanding:

- Clearly define the problem
- Identify goals: reduce crop loss, support farmers with timely intervention, improve yield
- Understand pain points: late disease detection, lack of expert access, manual diagnosis limitations
- Set success criteria: high model accuracy, early detection capability, ease of use in real-world conditions

2. Data Understanding:

- Explore the dataset(s): structure, image count, disease categories, labels, and sources
- Identify key data points: crop type, disease type, health status, image quality
- Look for patterns: common diseases per crop, visual symptoms, class imbalances
- Note external factors: crop season, weather conditions, growth stage

3. Data Preparation:

- Clean the data: remove poor-quality images, incorrect labels, duplicates
- Resize and normalize images to standard formats (e.g., 224x224, pixel scaling)
- Apply data augmentation (rotation, flip, zoom) to increase model generalization

4. Modeling:

- Choose model types: (e.g., ResNet, MobileNet, EfficientNet)
- Transfer learning using pretrained models
- Ensemble models for improved robustness
- Train the model with augmented data
- Tune hyperparameters (e.g., learning rate, batch size, epochs)

5. Evaluation:

- Use standard metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix
- Analyze misclassifications and confusion between diseases
- Compare different models and configurations
- Identify trade-offs between accuracy and speed (especially for field deployment)

6. Deployment/Delivery:

- Visualize detection results with image previews and disease labels
- Summarize key insights: most common diseases, detection accuracy per crop

- Suggest future improvements: Real-time detection from mobile camera, integration with weather or soil data for holistic farm advisory

GRADING GUIDELINE

No.	Name	Description	Grading Percentage
1	Data	Clearly defined data sources and collection methods	30%
2	AI/ML Technology	Explanation of AI model, techniques and implementation	40%
3	Demo	Functional code demonstrating the AI solution	20%
4	Reference & Citation	At least 5 credible sources supporting the project	10%